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Assessment Tool CO1
AT1
Application based

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① Training and validation loss

(i) Epoch at which overfitting begins

At Epoch 4, validation loss increases from 2.0 → 2.1, while training loss continues decreasing

Answer: overfitting begins at Epoch 4.

(ii) Two techniques to reduce overfitting

(1) Dropout: Randomly removes some neurons during training

(2) Regularization: Adds a penalty to large model weights and prevents the model from becoming too complex.

(iii) Effect on generalization

overfitting means the model learns the training data too well, including noise

- Training performance becomes very good
- performance on unseen data becomes poor
- Generalization ability decreases.

5. Training and validation accuracy

Epochs	Training accuracy	validation accuracy
1.	60%,	50%,
2.	65%,	62%,
3.	70%,	68%,
4.	75%,	72%,
5.	80%,	70%,

(i) Epoch at which overfitting starts.

At Epoch 3, training accuracy continues increasing but after Epoch 3, validation accuracy starts decreasing.

Here overfitting starts at Epoch 3

(ii) Why does validation accuracy decrease?

The model becomes too specialized to the training data. It learns training patterns and noise instead of general patterns.

- Training accuracy ↑
- Validation accuracy ↓
- performance on unseen data becomes poor

6. Medical Diagnosis system

- Total cases = 300
- TP = 138
- FN = 12
- TN = 135
- FP = 15

Formulas

Accuracy

$$= \frac{TP + TN}{TP + TN + FP + FN} \times 100 = \frac{138+135}{300} \times 100$$

$$= \frac{273}{300} \times 100 = 91$$

precision

$$\frac{TP}{TP+FP} \times 100 = \frac{138}{138+15} \times 100 = 90.26\%$$

Sensitivity

$$\frac{TP}{TP+FN} \times 100 = \frac{138}{138+12} \times 100 = 92\%.$$

Specificity

$$= \frac{TN}{TN+FP} \times 100 = \frac{135}{135+15} \times 100 = 90\%.$$

F1-Score.

$$= \frac{2(138)}{2(138)+15+12} \times 100$$

$$= \frac{276}{303} \times 100 = 91.09$$

3. Training and validation loss

Epoch	Training loss	validation loss
1.	3.0	2.7
2.	2.6	2.3
3.	2.3	2.1
4.	2.1	2.0
5.	1.9	2.2

(i) Epoch of optimal generalization

Ans:- optimal generalization occurs at Epoch 4

(i) Sign of overfitting

After Epoch 4:

- Training loss continues to decrease
- validation loss starts increasing

: Epoch 4 → validation loss = 2.0

. Epoch 8 → validation loss = 3.1

(ii) Two regularization techniques

1. L2 Regularization: penalizes large weights and makes the model simpler
2. Dropout: Randomly disables some neurons during training and reduces over-dependence on specific features.

Spam Email Filtering

TP = 180

FN = 20

TN = 170

FP = 30

Total = 400

1. Accuracy

$$= \frac{TP+TN}{Total} \times 100$$

$$= \frac{180+170}{400} \times 100$$

$$= 87.5$$

F_1 = Score

$$= \frac{2TP}{2TP+FP+FN} \times 100$$

$$= \frac{360}{360+50} \times 100$$

2. Precision

$$= \frac{TP}{TP+FP} \times 100$$

$$= \frac{180}{180+30} \times 100$$

3. Sensitivity

$$= \frac{TP}{TP+FN} \times 100$$

$$= \frac{180}{180+20} \times 100$$

$$= 90\%$$

4. Specificity

$$= \frac{TN}{TN+FP} \times 100 = \frac{170}{170+30} \times 100 = 85\%$$